# CD-900 Conveyor Drive: Service Manual

| Document | Revision | Effective | Owner |
|---|---|---|---|
| NG-MAN-CD900 | C | 2025-03-01 | Northgate Equipment Co. |

## 1. Overview

The Northgate CD-900 is a 10 HP (7.5 kW) gearmotor drive with variable frequency drive for belt and roller conveyors up to 300 ft. The drive reports motor temperature, power draw and gearbox vibration.

## 2. Specifications

| Item | Value |
|---|---|
| Motor | 7.5 kW, 460V / 3 ph, Class F insulation |
| Full load amps (FLA) | 11.8 A |
| Gearbox | Helical-bevel, 20:1 ratio |
| Belt speed | 60 to 240 ft/min |
| Thermal protection | PTC thermistor, trip at 155F frame temperature |

## 3. Normal Operating Ranges

| Measurement | Normal range | Alarm limit |
|---|---|---|
| Motor frame temperature | 100F to 135F | Above 155F (TH-4) |
| Power draw | 5.5 to 8.0 kW | 9.0 kW warning (OL-1), 10.5 kW for 30 s trip (OL-2) |
| Gearbox vibration | Below 3.5 mm/s | Above 6.0 mm/s |

## Fault Code Reference

| Code | Name | Trigger condition | Severity |
|---|---|---|---|
| OL-1 | Overload warning | Power draw above 9.0 kW for 60 seconds. | Low |
| OL-2 | Overload trip | Power draw above 10.5 kW for 30 seconds. VFD stops the motor. | High |
| JM-3 | Jam detected | Photo-eye blocked for more than 8 seconds while belt runs. | Medium |
| TH-4 | Motor thermal trip | Motor frame temperature above 155F. | High |

## Troubleshooting by Fault Code

### *OL-1: Overload warning*

**Trigger:** Power draw above 9.0 kW for 60 seconds.

**Likely causes, most common first:**

- Heavy product mix during peak hours.
- Belt over-tensioned.
- Dragging rollers.

**Corrective actions:**

1. Check the load on the conveyor.
2. Check belt tension (1% elongation target).
3. Spin rollers by hand under LOTO and replace any that drag.

### *OL-2: Overload trip*

**Trigger:** Power draw above 10.5 kW for 30 seconds. VFD stops the motor.

**Likely causes, most common first:**

- Product jam at a merge or transfer (most common during afternoon peak).
- Seized roller or bearing.
- Gearbox failure.
- Belt tracking off and rubbing the frame.

**Corrective actions:**

1. Apply LOTO before clearing any jam. Never reach into a stopped conveyor without LOTO.
2. Walk the full line and clear jams at merges and transfers.
3. Check for seized rollers and frame rub marks.
4. Let the motor cool below 135F before restart. Reset at the VFD keypad.
5. If OL-2 repeats 3 times in one shift, open a P2 work order and check the gearbox oil.

Repeated OL-2 trips also heat the motor. Check for TH-4 at the same time.

### *JM-3: Jam detected*

**Trigger:** Photo-eye blocked for more than 8 seconds while belt runs.

**Likely causes, most common first:**

- Product jam.
- Dirty photo-eye lens.

**Corrective actions:**

1. Apply LOTO and clear the jam.
2. Clean the photo-eye lens.

### *TH-4: Motor thermal trip*

**Trigger:** Motor frame temperature above 155F.

**Likely causes, most common first:**

- Repeated overloads.
- Blocked motor cooling fins.
- Low VFD output frequency for long periods.

**Corrective actions:**

1. Let the motor cool for at least 30 minutes.
2. Clean the cooling fins.
3. Resolve any OL-1 or OL-2 cause before restart.

## Preventive Maintenance

| Task | Interval |
|---|---|
| Check belt tension and tracking | Weekly |
| Inspect rollers | Monthly |
| Change gearbox oil | Every 10,000 hours or 2 years |
| Clean motor fins | Quarterly |

**WARNING:** Apply lockout/tagout per SOP-OPS-001 before opening any electrical enclosure or touching moving parts. Only authorized technicians may perform these steps.